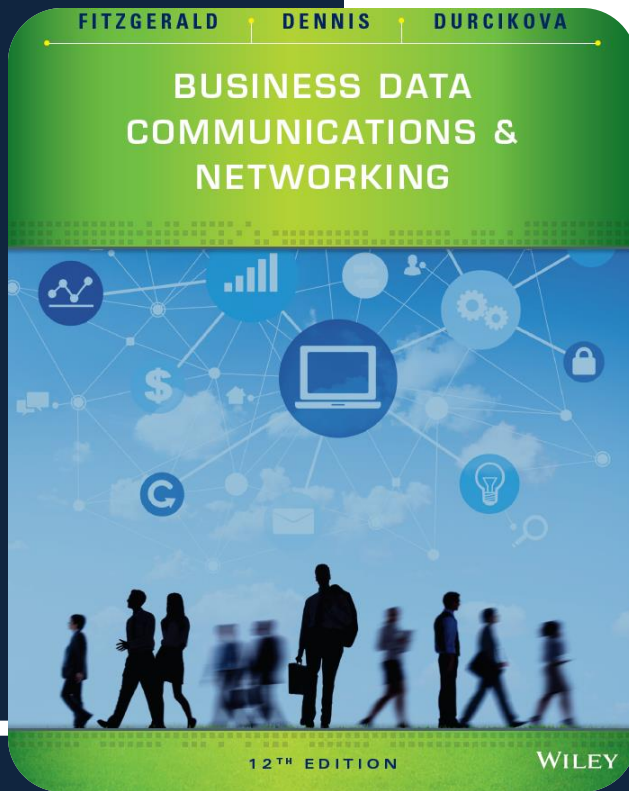


BUSINESS DATA COMMUNICATIONS & NETWORKING

Chapter 2 Application Layer



FitzGerald • Dennis • Durcikova

CHAPTER 2: APPLICATION LAYER

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Application Layer (Layer 7 of OSI Model, Layer 5 of Internet Model):

The **application layer** is the application **software** used by the **network user** and includes much of what the OSI model contains in the application, presentation, and session layers. It is the user's access to the network.

By using the application software, the user defines **what messages are sent** over the network.

Because, it is the layer that most people understand best and because starting at the top sometimes helps people understand better.

It discusses the architecture of network applications and several types of network application software and the types of messages they generate.

- Application Architectures
- The Web (WWW)
- Email
- Other Application-Layer Protocols
- Implications for Management

Application Layer

- Layer 5 in the Internet model
- The software that enables **users** to interact with the network and accomplish tasks

Internet Model

Application

Transport

Network

Data Link

Physical

Application Architecture

- The way the functions of the application layer are spread out across the client and server
- Four components of applications:
 1. Presentation logic
 2. Application logic
 3. Data access logic
 4. Data Storage

In different application architectures, these components are in different locations (client or server-host)

Host-Based Architecture

- **Server contains all components** (“server-based”)
- Common in the 1960s with mainframes and terminals

CLIENT/Terminal



Processing, Data

SERVER/Host



Presentation Logic
Application Logic
Data Access Logic
Data Storage

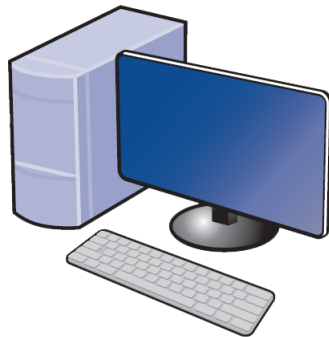
Host-Based Architecture

- Advantages
 - Very simple (centralized installation)
 - Single point of control (centralized security)
- Disadvantages
 - Host (server) can become a bottleneck (processing problems)
 - Upgrades typically expensive

Client-Based Architecture

- Client contains presentation, application, and data access logic
- Most common in the 1980s

Processing
CLIENT



Presentation Logic
Application Logic
Data Access Logic

Data
SERVER



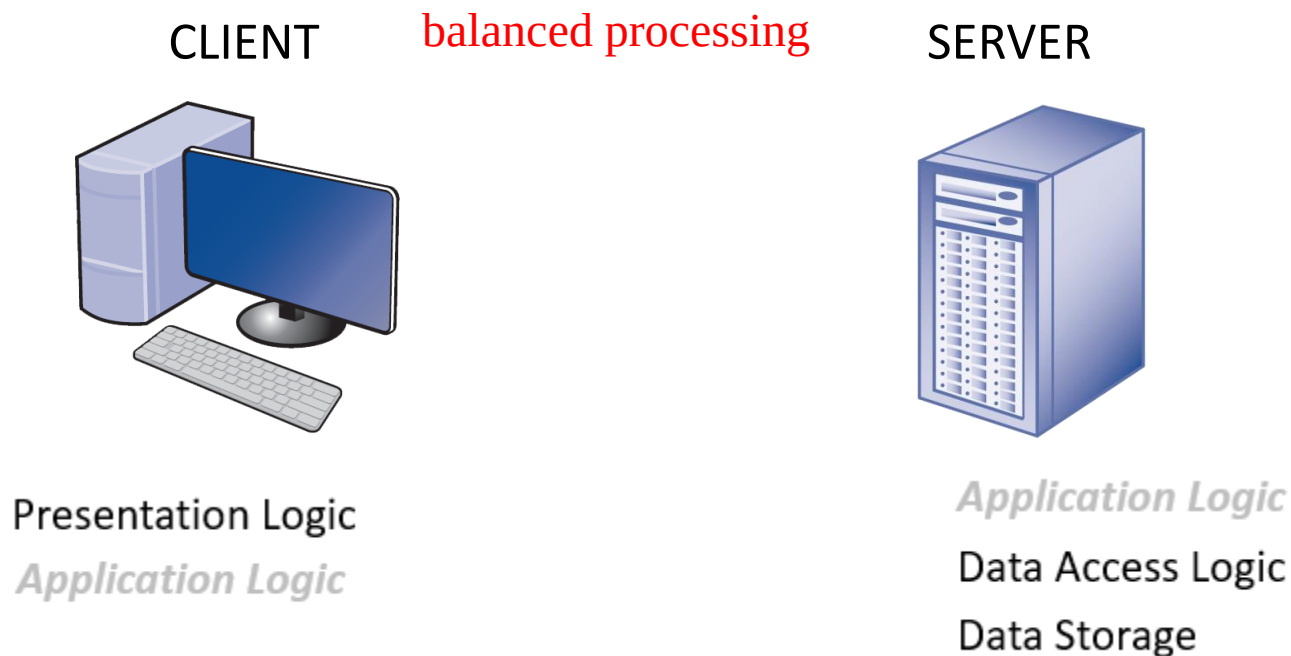
Data Storage

Client-Based Architecture

- Advantages
 - Hardware and applications **less expensive** (multiple vendors for software and hardware)
 - **Simple** architecture
- Disadvantages
 - **Data must travel back and forth** between server and client

Client-Server Architecture

- Most common architecture today



Client-Server Architecture

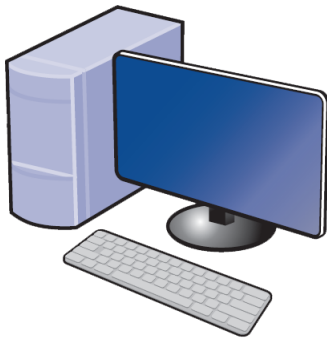
More functionality

Thick-Client Architecture



CLIENT

SERVER



Presentation Logic
Application Logic

Data Access Logic
Data Storage

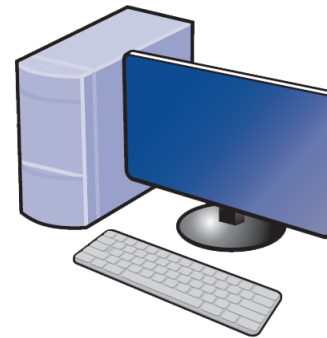
Easier to manage

Thin-Client Architecture



CLIENT

SERVER



Presentation Logic

Application Logic
Data Access Logic
Data Storage

Development and maintenance costs
can be higher than for thin clients.

Client-Server Architecture

- Advantages
 - More efficient because of distributed processing
 - Allows hardware/software from **different vendors** to be used together (**less expensive**)
 - **Less bandwidth** required
- Disadvantages
 - May be challenges in **configuring hardware/software from different vendors to work together**
 - In many cases, **middleware** is required

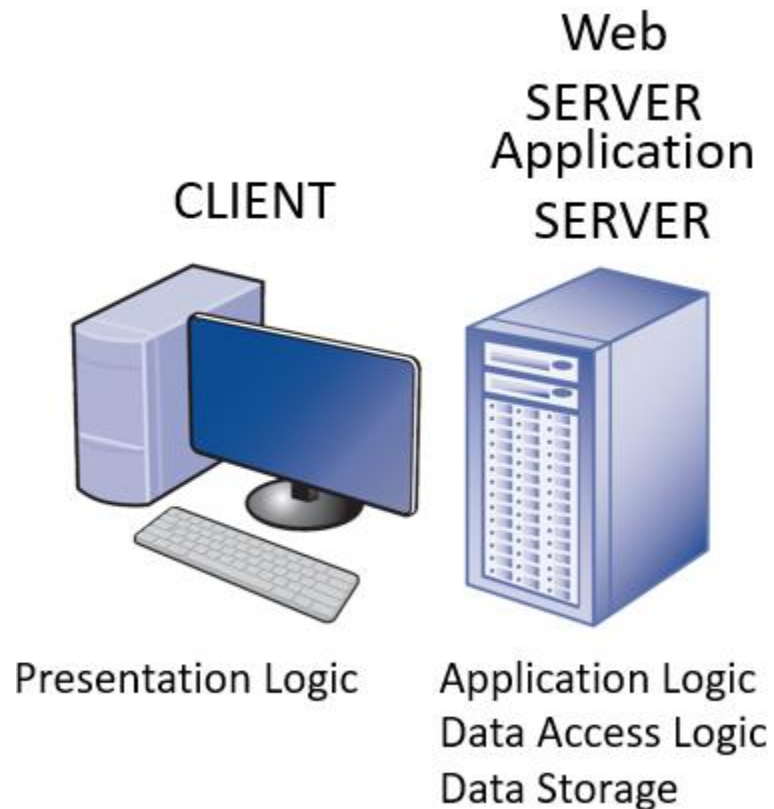
Client-Server Architecture

- **Middleware** is software acts as an intermediary by “sitting between” client and server applications
- Provides a standard way of translating between software from different vendors
- Manages message transfers
- Insulates network changes from the clients (e.g., adding a new server)



Client-Server Architecture

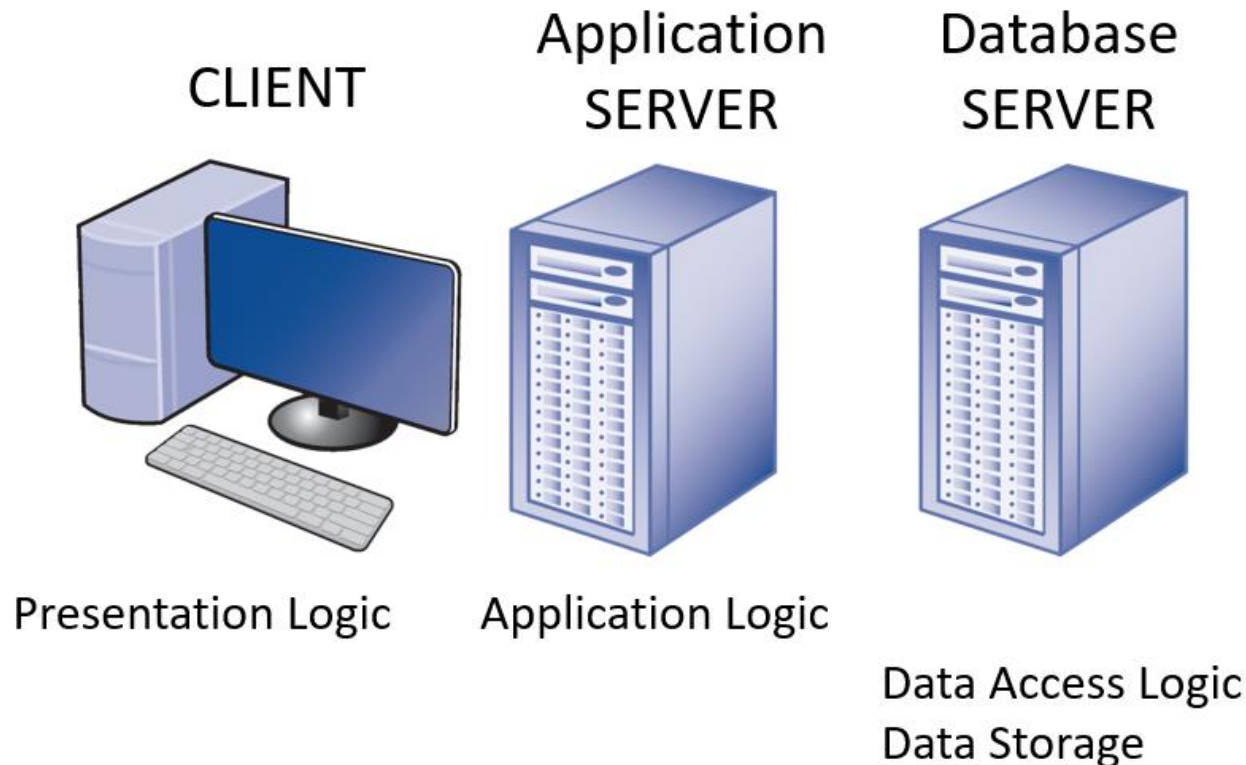
- Example of **two-tier** architecture



Two-tiered architectures have only clients and servers

Client-Server Architecture

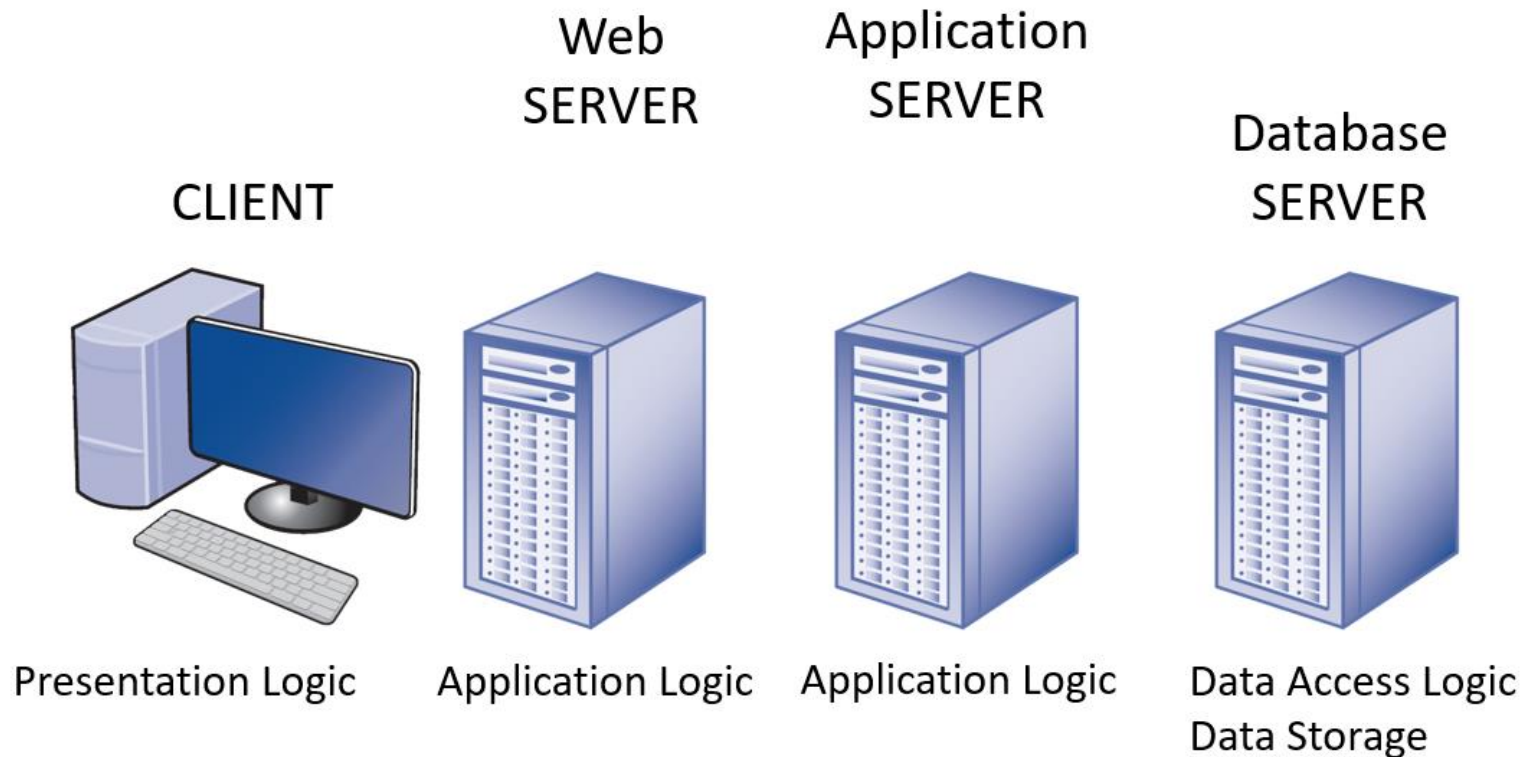
- Example of **three-tier** architecture



(1) presentation logic, (2) application logic, and (3) and data access logic and storage are **separated**.

Client-Server Architecture

- Example of **n-tier** architecture



More than one tier may be used to support application logic, typically due to a Web server tier being included

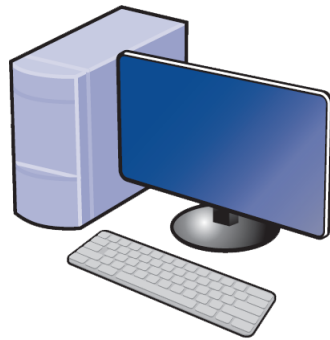
Tiered Client-Server Architecture

- Advantages
 - Load **balancing** (number of servers are increased)
 - More scalable
- Disadvantages
 - Each tier increases network load (**more data transfer**)
 - More **complex** and difficult to develop applications

Peer-to-Peer (P2P) Architecture

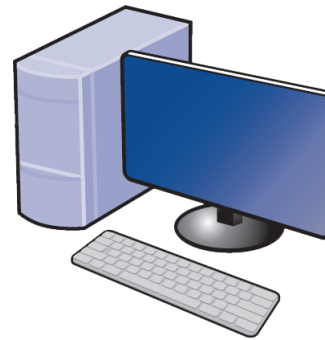
- An older architecture that became popular again with Napster, BitTorrent, etc.
- All devices can **act as client and server**

CLIENT/SERVER



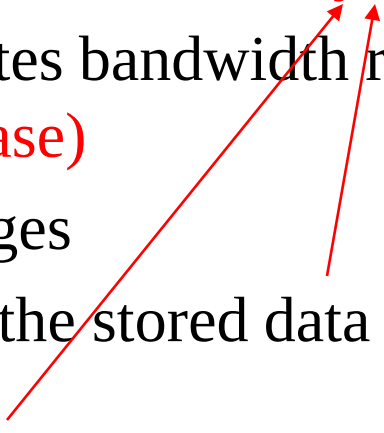
Presentation Logic
Application Logic
Data Access Logic
Data Storage

CLIENT/SERVER



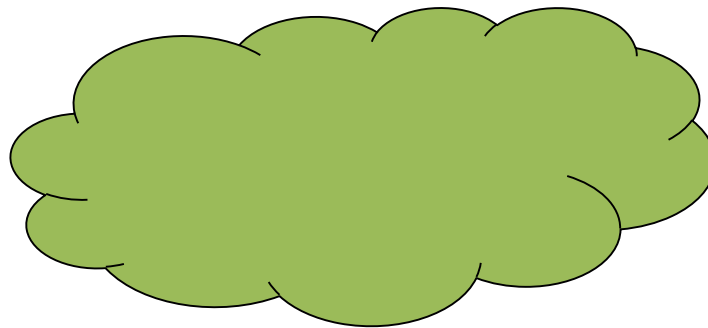
Presentation Logic
Application Logic
Data Access Logic
Data Storage

Peer-to-Peer Architecture

- Advantages
 - Resilient to failure
 - Data can be stored anywhere on network
 - Distributes bandwidth requirements (compared to one server case)
 - Disadvantages
 - Finding the stored data
 - Security
- 

Cloud Computing

- **Cloud Computing** is the general term for **enabling access to computing services** over the network (most commonly the Internet)
- Models of cloud computing define who manages each application function and associated hardware/software



Cloud Computing

- Cloud Computing Models
 - **Infrastructure as a Service (IaaS)**
 - All hardware is outsourced
 - **Platform as a Service (PaaS)**
 - Application logic and data are managed internally
 - e.g., Microsoft Azure
 - **Software as a Service (SaaS)**
 - All application components and associated hardware/software outsourced
 - Based on multitenancy
 - e.g. Salesforce.com

Cloud Computing

	Traditional Thin-Client Client-Server		Infrastructure as a Service (IaaS)		Platform as a Service (PaaS)		Software as a Service (SaaS)	
	Internal	Outsourced	Internal	Outsourced	Internal	Outsourced	Internal	Outsourced
Application Logic	X		X		X			X
Data Storage	X		X		X			X
Data Access Logic	X		X			X		X
Operating System	X		X			X		X
Virtualization Software	X		X			X		X
Server Hardware	X			X		X		X
Storage Hardware	X			X		X		X
Network Hardware	X			X		X		X

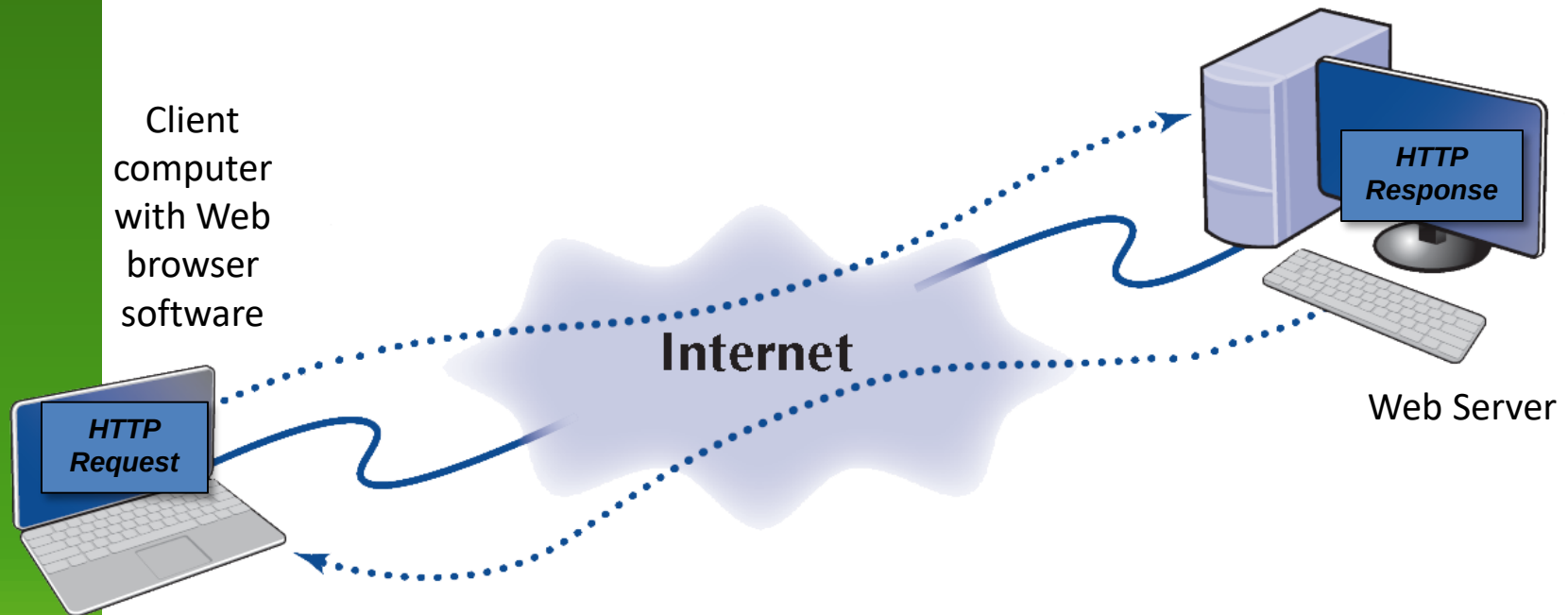
Criteria for Choosing Architecture

- Infrastructure (hardware)
 - Cost of servers, clients, and circuits
 - Reliability
- Development Costs
 - Software; cheaper on host-based architectures
- Scalability
 - Ability to increase (or decrease) in computing capacity as network demand changes
 - Easier in client-server architectures

The Web

- Hypertext Transfer Protocol (HTTP)
- HTTP Request and Response

The standard protocol for communication between a Web browser and a Web server

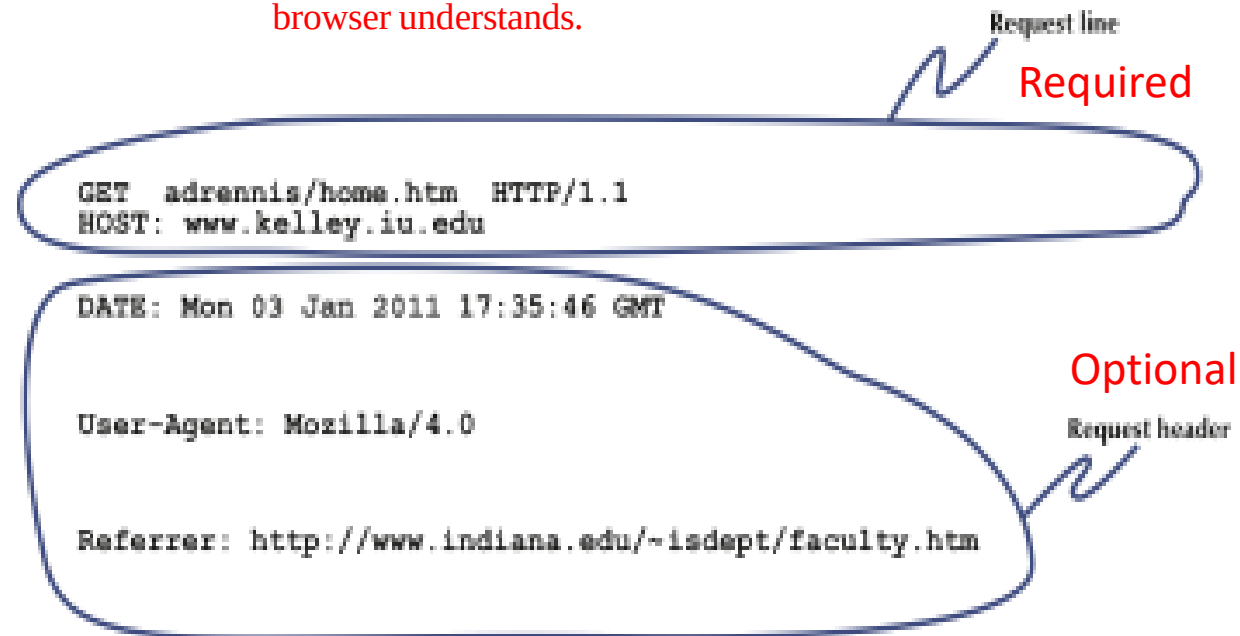


- HTTP Request

Request line, starts with a command (e.g., GET), provides the URL, and ends with the HTTP version number that the browser understands.

FIGURE 2-11

An example of a request from a Web browser to a Web server using the HTTP (Hypertext Transfer Protocol) standard



Request header contains a variety of optional information such as the Web browser being used (e.g., Internet Explorer), the date, and a userid and password for use if the Web page is password-protected.

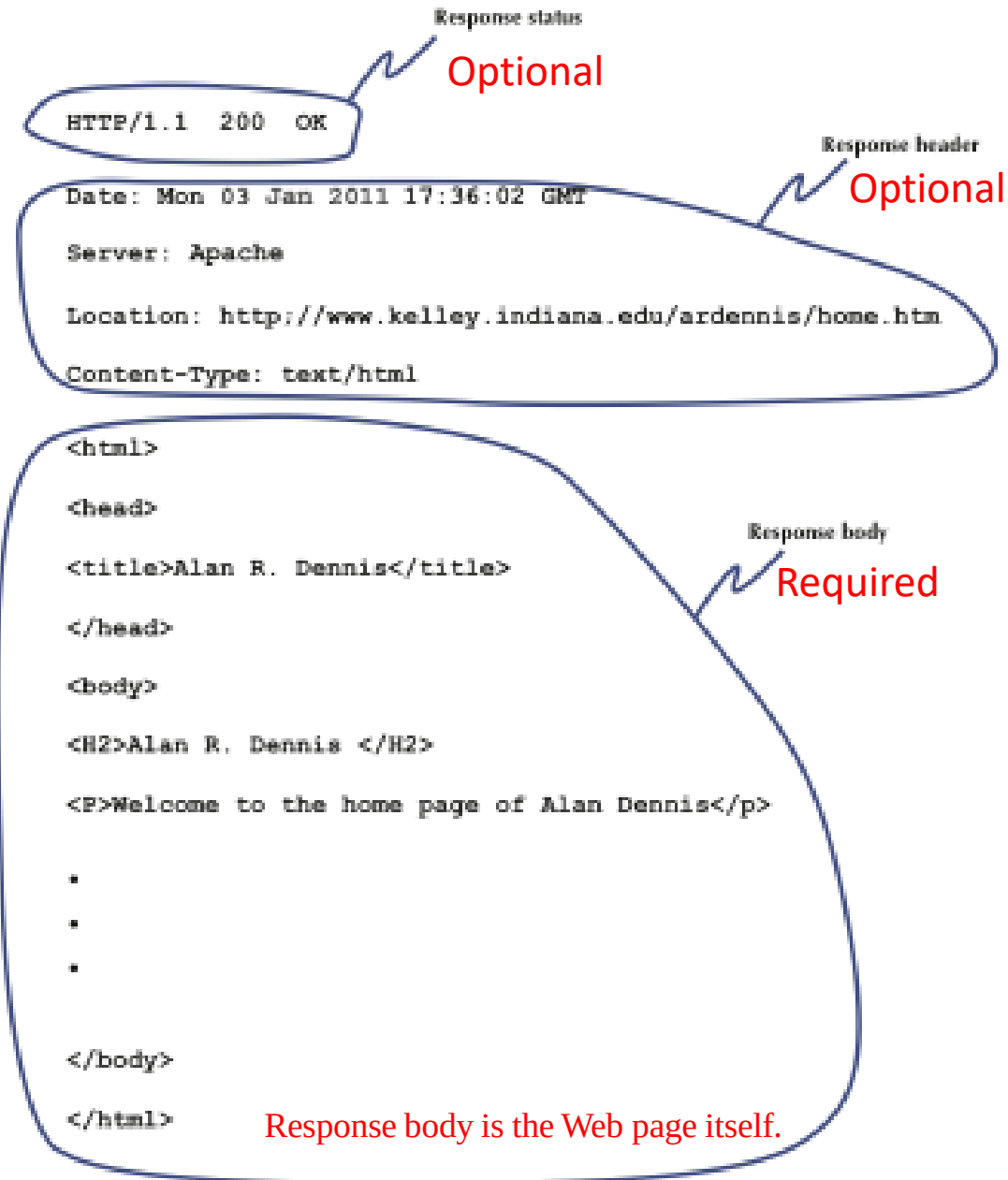
contains the HTTP version number the server has used, a status code (e.g., 200 means OK, 404 means page not found)

The Web

- HTTP Response

FIGURE 2-12

An example of a response from a Web server to a Web browser using the HTTP standard



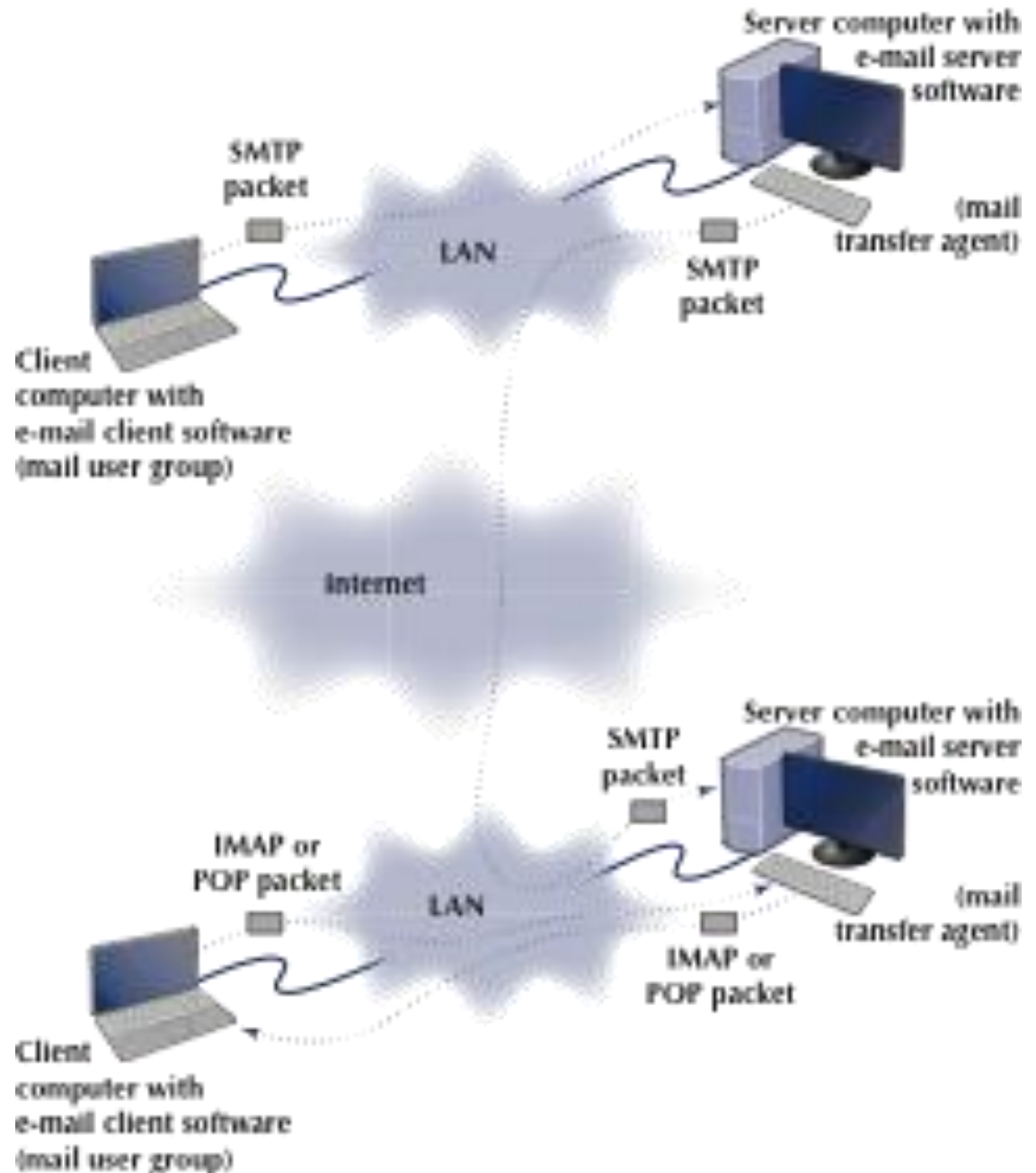
Email

- **Mail Transfer Agent (MTA)**
 - Formal name for mail server software
 - e.g., Sendmail, Postfix,
- **Mail User Agent (MUA)**
 - Formal name for mail client software
 - e.g., Outlook, Apple Mail, Thunderbird
- **Simple Mail Transfer Protocol (SMTP)**
 - Protocol used to send a message to a MTA (client to server)
 - Originally only handled text files
- **Internet Message Access Protocol (IMAP) or Post Office Protocol (POP)**
 - Protocols used by a MUA to retrieve messages from an MTA (server to client)
- **American Standard Code for Information Interchange (ASCII)**
 - A standard for encoding text characters (a-z, A-Z, 0-9, a few symbols)

Email

FIGURE 2-13

How SMTP (Simple Mail Transfer Protocol) email works. IMAP = Internet Message Access Protocol; LAN = local area network



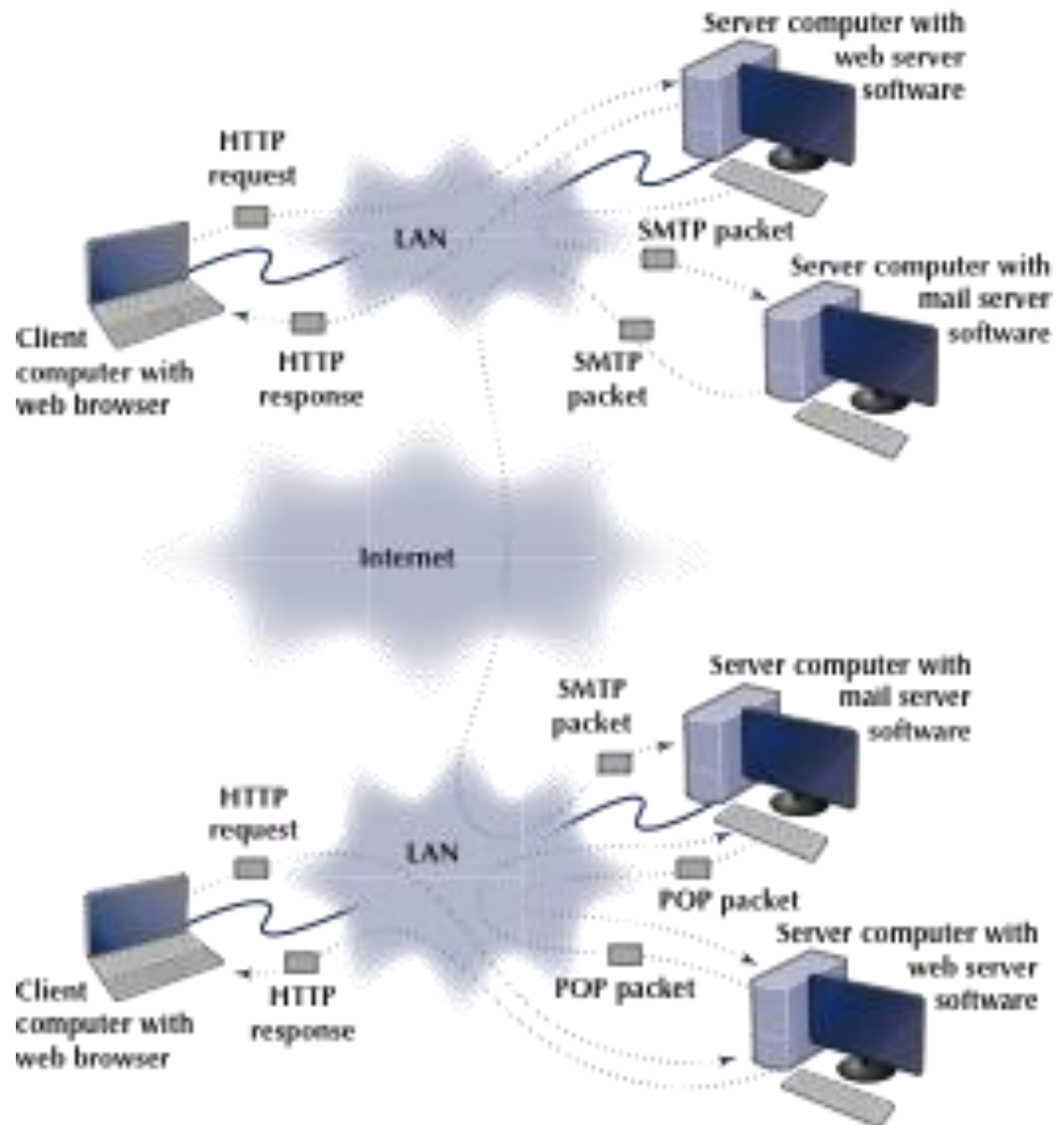
Email

- How a message is sent (2-tier, thick client)
- Sending Client → Sender's Mail Server (SMTP)
- Sender's Mail Server → Receiver's Mail Server (SMTP)
- Message waits on Receiver's Mail Server in “mailbox”
- Receiving Client → Receiver's Mail Server (IMAP or POP)
- Receiver's Mail Server → Receiving Client (IMAP or POP)

Email

FIGURE 2-14

Inside the Web. HTTP = Hypertext Transfer Protocol; IMAP = Internet Message Access Protocol; LAN = local area network; SMTP = Simple Mail Transfer Protocol

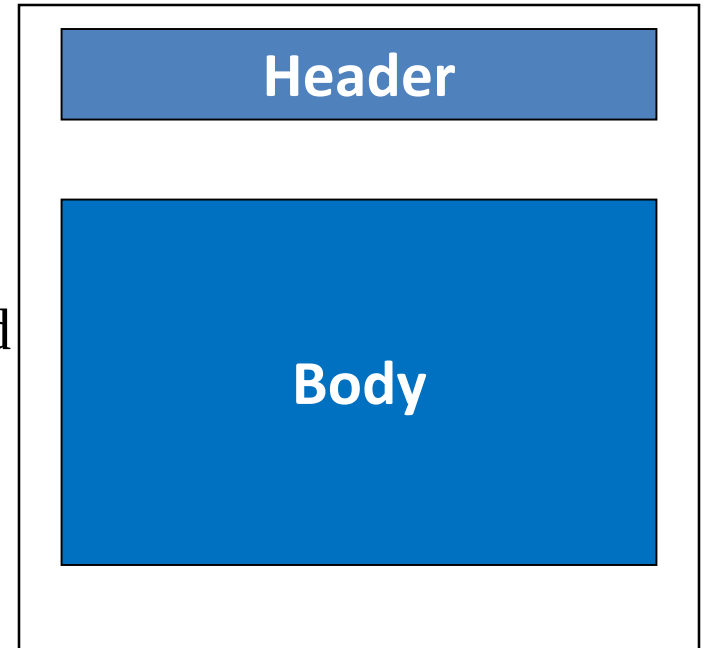


Email

- How a message is sent via webmail (3-tier, thin client)
 1. Sending Client → Sender's Web Server (**HTTP**)
 2. Sender's Web Server → Sender's Mail Server (**SMTP**)
 3. Sender's Mail Server → Receiver's Mail Server (**SMTP**)
 4. Mail waits on Receiver's Mail Server in “mailbox”
 5. Receiving Client → Receiver's Web Server (**HTTP**)
 6. Receiver's Web Server → Receiver's Mail Server (**IMAP or POP**)
 7. Receiver's Mail Server → Receiver's Web Server (**IMAP or POP**)
 8. Receiver's Web Server → Receiving Client (**HTTP**)

Email

- SMTP Message Format
 - RFC 822: Standard for text message format
- Header lines
 - Contain information about the message such as To, From, and Subject
- Body section
 - Contains the “content of the message”
 - Begins with the “DATA” keyword
 - Only uses ASCII characters



Email

- Inside an SMTP packet
 - **Header:** source, destination, date, subject, etc.
 - **Body:** keyword “DATA”, then email message

FIGURE 2-15

An example of an email message using the SMTP (Simple Mail Transfer Protocol) standard

The diagram illustrates the structure of an SMTP packet. It consists of two main parts: a header and a body. The header is enclosed in a blue oval and contains the following text: FROM: "Alan Dennis" <cardennis@indiana.edu>, TO: "Pat Someone" <someone@somewhere.com>, DATE: Mon 03 Jan 2011 19:03:03 GMT, SUBJECT: Sample Note, and Message-ID: <4.1.20000623164823.009f5e808IMAP.IU.EDU>. A blue arrow points from the word "Header" to this oval. The body is enclosed in a blue oval and contains the text "DATA" followed by "This is an example of an e-mail message.". A blue arrow points from the word "Body" to this oval.

```
FROM: "Alan Dennis" <cardennis@indiana.edu>
TO: "Pat Someone" <someone@somewhere.com>
DATE: Mon 03 Jan 2011 19:03:03 GMT
SUBJECT: Sample Note
Message-ID: <4.1.20000623164823.009f5e808IMAP.IU.EDU>

DATA
This is an example of an e-mail message.
```

Email

- SMTP is a simple protocol to send plain text
- Not designed to send images or attachments
- **Multipurpose Internet Mail Extension (MIME)**
 - A standard to extend support for attachments and non-ASCII characters in email
 - Used by sender to convert (encode) any non-ASCII content into ASCII
 - Receivers then convert (decode) the ASCII back to its original format

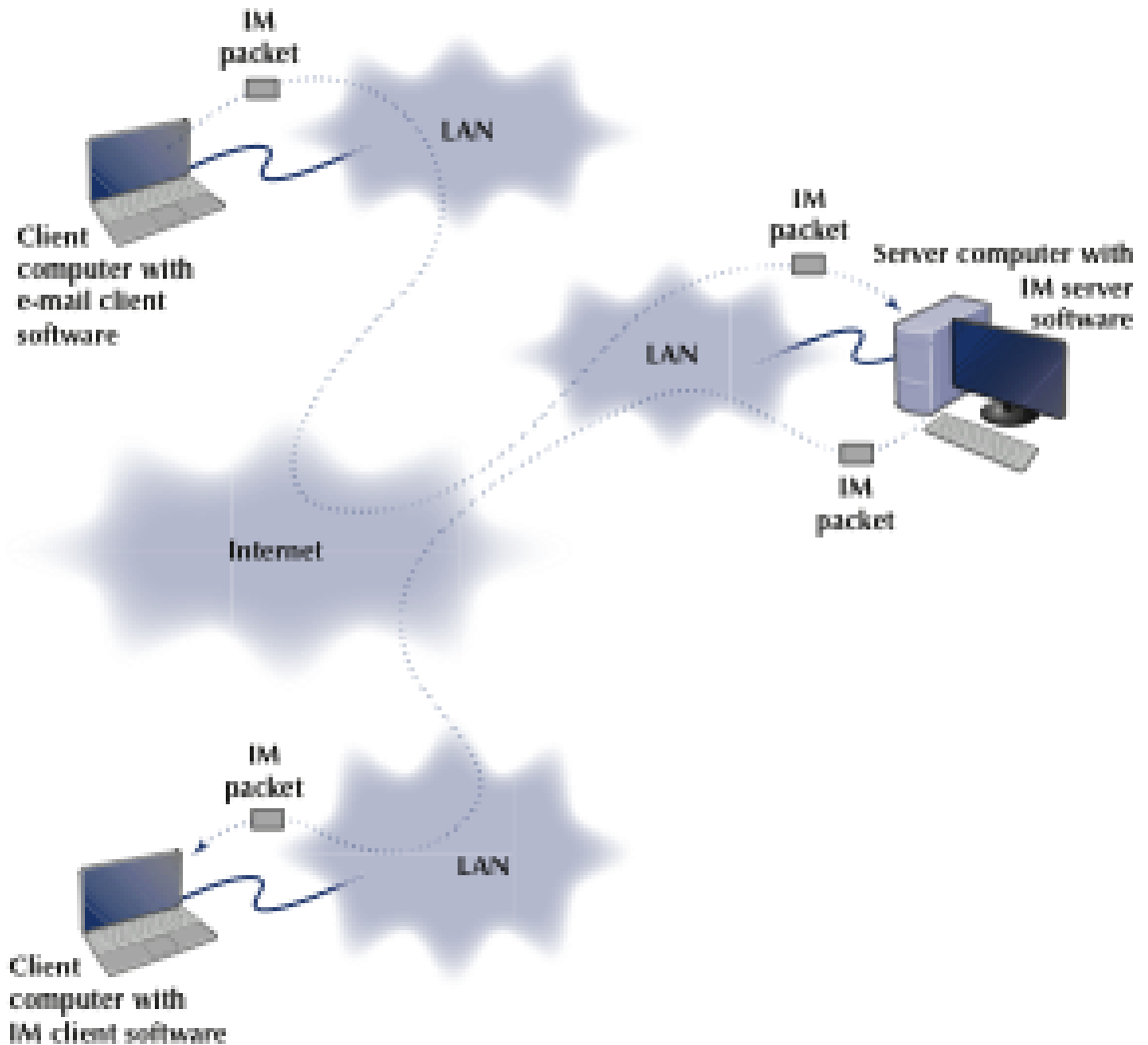
Other Application-Layer Protocols

- **File Transfer Protocol (FTP)** - for moving files between clients and servers
- **Telnet, Secure Shell (SSH)** - for executing commands on a remote system
- **Internet Relay Chat (IRC), Extensible Messaging and Presence Protocol (XMPP)** - for real-time text chat (instant messaging)
- **Domain Name System (DNS)** - for mapping domain names to IP addresses

Instant Messaging

FIGURE 2-16

How instant messaging (IM) works. LAN = local area network



Video Conferencing

FIGURE 2-17

A Cisco telepresence system

Source: Courtesy Cisco Systems, Inc. Unauthorized use not permitted



Implications for Management

- Primary purpose of networks is to provide environment for applications
- The number and types of applications on the network is dramatically increasing